

Brownian Motion and Its Connections to PDEs

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Introduction

This document explores the deep connections between Brownian motion and partial differential equations (PDEs), focusing on four key topics:

1. Brownian Motion and the Heat Equation
2. Itô's Lemma and PDEs
3. The Feynman-Kac Formula
4. The Dirichlet Problem and Brownian Motion

Each section includes rigorous derivations, comments on intuition, and transitions between concepts.

1 Brownian Motion and the Heat Equation

Mathematical Setup

Let B_t be a standard Brownian motion in $\mathbb{R}^d$. Its transition density $p(t, x)$ represents the probability density of B_t starting at $x = 0$:

$$p(t, x) = \frac{1}{(2\pi t)^{d/2}} \exp\left(-\frac{|x|^2}{2t}\right).$$

Connection to the Heat Equation

The density $p(t, x)$ satisfies the **heat equation**:

$$\frac{\partial p}{\partial t} = \frac{1}{2} \Delta p, \quad \text{where } \Delta = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2}.$$

Verification:

$$\begin{aligned}\frac{\partial p}{\partial t} &= -\frac{d}{2t}p + \frac{|x|^2}{2t^2}p, \\ \Delta p &= \sum_{i=1}^d \left(\frac{x_i^2}{t^2} - \frac{1}{t} \right) p = \left(\frac{|x|^2}{t^2} - \frac{d}{t} \right) p.\end{aligned}$$

Substituting into the heat equation confirms the equality.

Comments

- The heat equation describes how the probability density of B_t diffuses over time.
- The factor $\frac{1}{2}$ arises from the variance of Brownian motion: $\text{Var}(B_t) = t$.

2 Itô's Lemma and PDEs

Itô's Lemma (1D Case)

Let X_t be an Itô process:

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dB_t.$$

For a $C^{1,2}$ function $u(t, x)$, Itô's Lemma states:

$$du(t, X_t) = \left(\frac{\partial u}{\partial t} + \mu \frac{\partial u}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 u}{\partial x^2} \right) dt + \sigma \frac{\partial u}{\partial x} dB_t.$$

Link to PDEs

Taking expectations and setting the drift term to zero (for martingales) yields the **Kolmogorov backward equation**:

$$\frac{\partial u}{\partial t} + \mu \frac{\partial u}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 u}{\partial x^2} = 0.$$

Example: Black-Scholes PDE

For a stock price S_t with $dS_t = rS_t dt + \sigma S_t dB_t$, the option price $V(t, S_t)$ satisfies:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{\sigma^2 S^2}{2} \frac{\partial^2 V}{\partial S^2} - rV = 0.$$

Comments

- Itô's Lemma is the stochastic analog of the chain rule, with an extra second-order term due to quadratic variation.
- The PDE governs expectations of the form $\mathbb{E}[u(T, X_T) | X_t = x]$.

3 The Feynman-Kac Formula

Statement

The solution to the PDE:

$$\frac{\partial u}{\partial t} + \mu \frac{\partial u}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 u}{\partial x^2} + f(t, x) - V(t, x)u = 0, \quad u(T, x) = \psi(x),$$

is given by:

$$u(t, x) = \mathbb{E} \left[\int_t^T e^{-\int_t^s V(r, X_r) dr} f(s, X_s) ds + e^{-\int_t^T V(r, X_r) dr} \psi(X_T) \middle| X_t = x \right].$$

Special Case: Heat Kernel

If $\mu = 0$, $\sigma = 1$, $f = V = 0$, then:

$$u(t, x) = \mathbb{E}[\psi(B_T) | B_t = x],$$

which solves $\frac{\partial u}{\partial t} + \frac{1}{2} \Delta u = 0$.

Comments

- Feynman-Kac provides a probabilistic representation of PDE solutions.
- Widely used in finance for pricing derivatives (e.g., zero-coupon bonds).

4 The Dirichlet Problem and Brownian Motion

Problem Statement

Solve $\Delta u = 0$ in a domain $D \subset \mathbb{R}^d$ with boundary condition $u = \phi$ on ∂D .

Probabilistic Solution

Let $\tau = \inf\{t \geq 0 : B_t \notin D\}$. The solution is:

$$u(x) = \mathbb{E}[\phi(B_\tau) | B_0 = x].$$

Example: Harmonic Functions in a Ball

For $D = B(0, R)$ and ϕ continuous:

$$u(x) = \int_{\partial B(0, R)} \phi(y) \frac{R^2 - |x|^2}{R|y - x|^d} d\sigma(y).$$

Comments

- Brownian motion "probes" the boundary to solve the PDE.
- Connects potential theory to stochastic processes.

Conclusion

These topics illustrate the rich interplay between Brownian motion and PDEs, with applications ranging from physics to mathematical finance. Key takeaways:

- Brownian motion provides a probabilistic interpretation of diffusion (heat equation).
- Itô's Lemma links stochastic calculus to PDEs via martingales.
- Feynman-Kac and the Dirichlet problem offer Monte Carlo methods for PDE solutions.